INCH-POUND

MIL-DTL-22520/40
16 June 2008

DETAIL SPECIFICATION SHEET

CRIMPING TOOLS, TERMINAL, HAND, WIRE TERMINATION; TYPE II T-HEAD FOR TERMINAL LUGS AND SPLICES SIZE 22-14

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-22520.

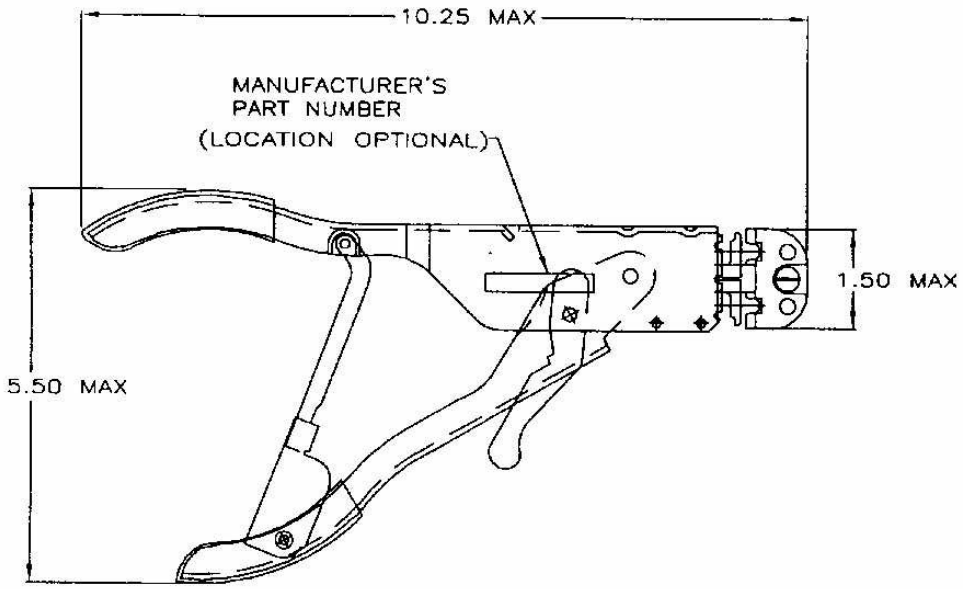


FIGURE 1. Crimp tool configuration.

AMSC N/A FSC 5120

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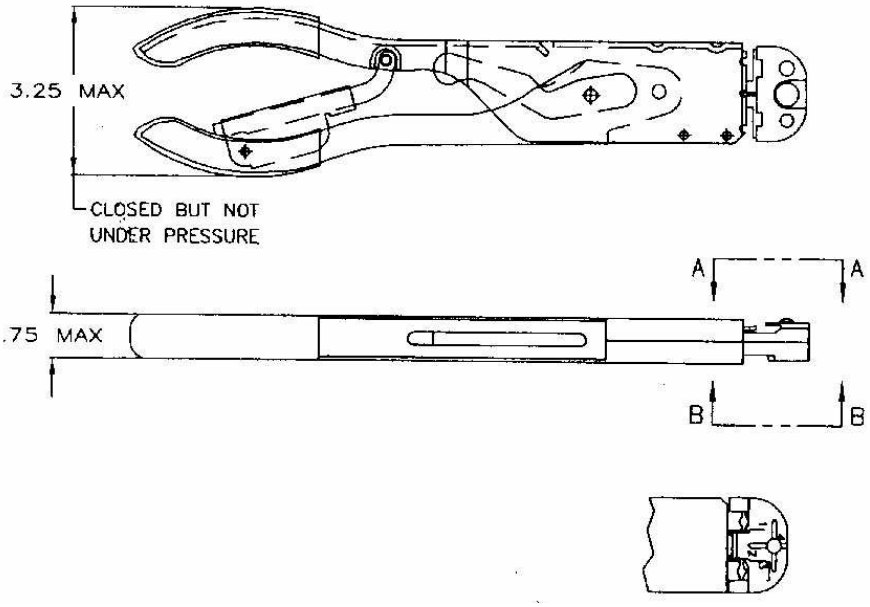


FIGURE 1. Crimp tool configuration - (Continued).

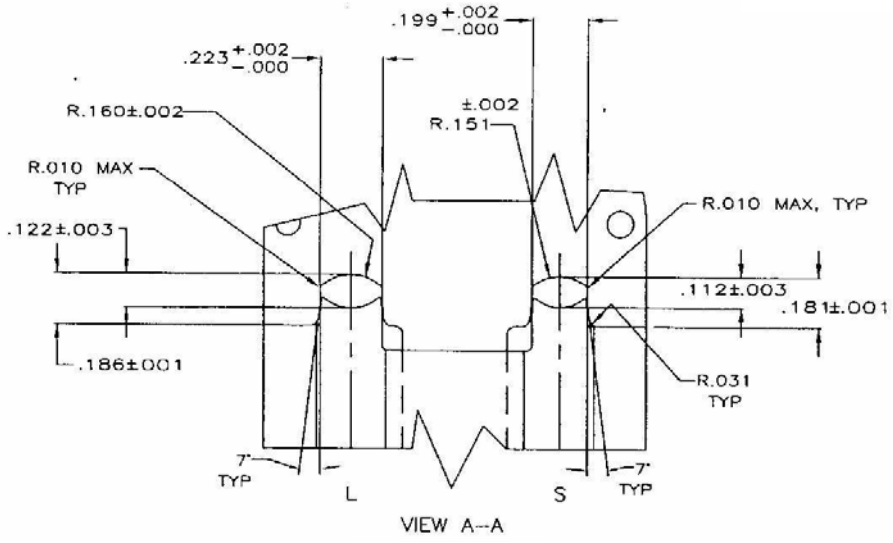


FIGURE 2. Wire barrel crimp dimensions.

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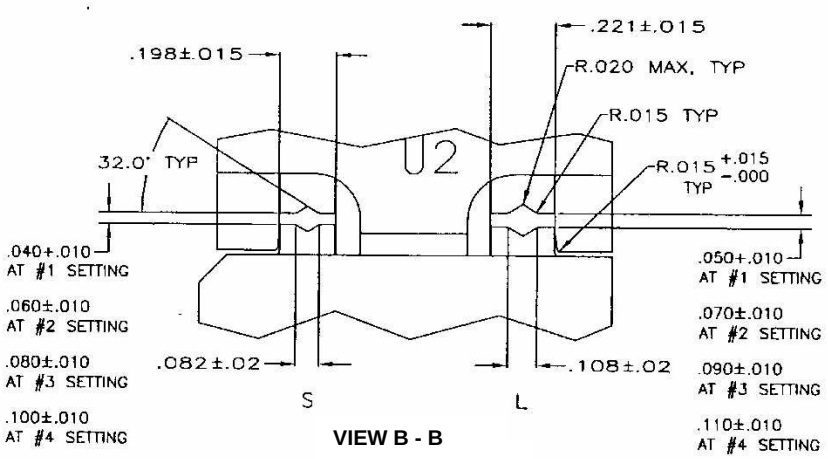


FIGURE 3. Insulation barrel crimp dimensions.

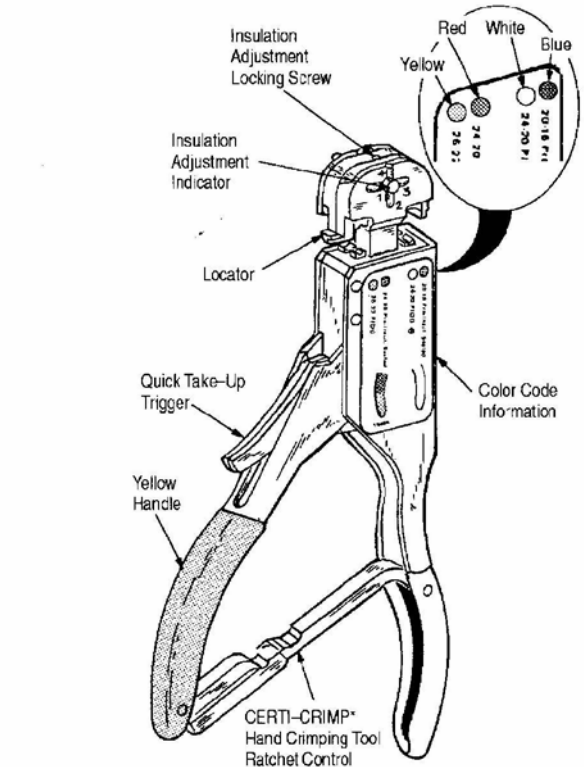


FIGURE 4. MIL-DTL-22520/40 Crimping tool showing four adjustment settings.

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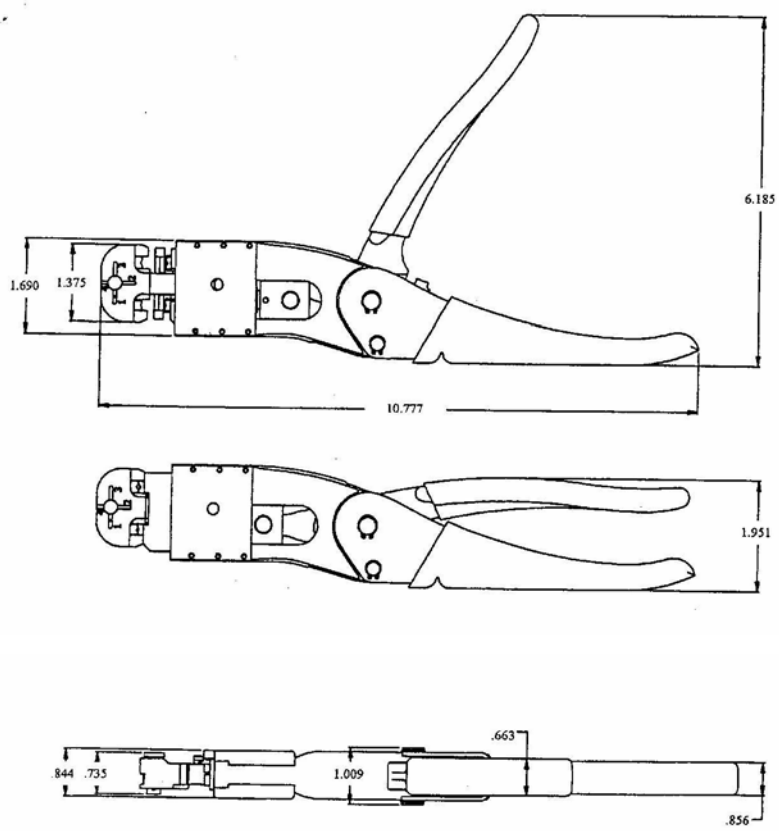


FIGURE 5. Crimp tool optional design.

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TABLE I. Available gages for tool.

Gage part number	Gage type	Wire size (AWG)	Gage area
MIL-DTL-22520/43-03	Go/No-Go	22-16	Insulation (position #1)
MIL-DTL-22520/43-04	Go/No-Go	16-14	Insulation (position #1)
MIL-DTL-22520/43-01	Go/No-Go	22-16	Wire
MIL-DTL-22520/43-02	Go/No-Go	16-14	Wire

TABLE II. Metric equivalents.

Inches	MM
0.001	0.025
0.002	0.051
0.010	0.254
0.013	0.330
0.015	0.381
0.020	0.508
0.031	0.787

Inches	MM
0.040	1.016
0.050	1.270
0.060	1.524
0.070	1.778
0.080	2.032
0.082	2.083
0.090	2.286
0.100	2.540
0.110	2.794
0.112	2.84

Inches	MM
0.122	3.099
0.151	3.835
0.160	4.064
0.181	4.597
0.186	4.724
0.197	5.004
0.198	5.029
0.199	5.055

Inches	MM
0.221	5.613
0.223	5.664
0.224	5.690
0.663	16.840
0.75	19.050
0.735	18.669
0.844	21.438
0.856	21.742
1.099	27.915
3.375	34.925
1.50	38.100
1.690	42.926
1.951	49.556
3.25	82.550
5.50	139.700
6.185	157.100
10.25	260.350
10.777	273.736

NOTES:

1. Dimensions are in inches.
2. Metric equivalents when specified are based upon 1.00 inch = 25.4 millimeters.
3. Unless otherwise specified herein or in MIL-DTL-22520, tolerances are ± .010 inch (0.25 millimeters) for three place decimals and ± .5° for angles.
4. Refer to MIL-DTL-22520/43 for calibration, inspection gage (M22520/43-01 through -04).

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REQUIREMENTS:

1. The envelope dimensions of the tool shall be in accordance with figure 1 or figure 5. Outline configuration optional within maximum dimensions shown for closed handle position.
2. The wire barrel crimp gaging dimensions and insulation barrel crimp gaging dimensions shall be in accordance with figures 2 and 3 when the tool is in the closed handle position.
3. The tool weight shall not be greater than 2.25 lb (1.02 kg).
4. Material: Jaw - 38 point carbon min. alloy steel Heat Treat Rockwell C 45 – 50 or equivalent.
   Handles - Steel or equivalent.
   Grips - Plastisol or equivalent (color optional).
5. Finish: Black oxide or equivalent.
6. Cycle control (ratchet) adjustment device shall be set to release between 45 to 60 pounds and shall be tamper proof.
7. Convex crimp surfaces shall be polished to 16 (RMS) Microfinish minimum.
8. Each tool shall have an instruction sheet which includes applications, usage and instructions.
9. Tool part number: M22520/40-01
10. Complete manufacturer's name or AIR-1351 logo shall be permanently marked on the tool.
11. The tool shall have four (4) adjustment settings to regulate the insulation crimp height (see figure 4). Pin in shaft mechanism (as shown on figure 4) or selector knob, or other mechanisms are permitted as long as they achieve the same functional adjustment/latch, and the adjustment mechanism is user friendly.
12. Qualification testing shall be in accordance with MIL-DTL-22520 type 2 tool requirements. Initial qualification test samples shall consist of a sample lot of terminals and/or splices from all sources (brands) qualified at the time of tool testing, and the tool is required to meet the minimum requirements with all samples tested.
13. Locator (positioner) design optional providing it meets the performance requirements of all terminals/splices used with the tool.
14. Optional design shall have the same marking requirements as shown on figure 3.

APPLICATION:

The S cavity is for crimping the wire barrel of SAE AS17143, SAE AS25036, terminals and splices in sizes 22 - 16 and cavity L is for sizes 16 - 14.

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CONCLUDING MATERIAL

Custodians:
Navy - AS
Army - CR
AF - 11

Preparing activity:
Navy - AS

(Project 5120-2007-001)

Review activities:
Navy - EC, MC, OS, SH
DLA - CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST online database at http://assist.daps.dla.mil.

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